Collision Theory

Name: ______________________ Class: ______________ Date: ______________

Part A

A1 A collision between two reactant molecules will only produce a reaction if certain conditions are met. State the **two** conditions the simulator tests for every collision.

A2 Define **activation energy** (E_{a}).

A3 The simulator reports a percentage called *Effective collisions*. In terms of collision theory, explain exactly what this percentage is measuring.

A4 A student claims: "*Whenever two reactant molecules touch, they react.*" Using what the simulator shows, explain why this claim is wrong.

Part B

Select $CO + NO_2 \longrightarrow CO_2 + NO$. **Turn OFF** "Show reaction conditions" and set the **Collision energy to maximum**.

B1 Using single **Collide** shots, try the angles below and record whether each one reacts.

Angle of CO	0°	90°	180°	your choice: _______
Reaction? (✓/×)				

B2 In the boxes, **draw** the molecules at the instant of collision for a successful hit and for a collision that just bounces. Label the **carbon atom** of CO in each.

Reacts	**Bounces**

B3 Explain *why* the carbon end of CO must be the part that strikes an oxygen of NO_2. (Hint: which new bond has to form?)

B4 Now press **Run ×50** (still at maximum energy). Record the *Effective collisions* value: ____________%. Then select $Cl + O_3 \longrightarrow ClO + O_2$ (a lone chlorine radical, $Cl^{\bullet}$), keep energy at maximum, and **Run ×50**: ____________%.

The two effective-collision percentages are very different. Explain what causes the gap, and describe the general method you could use to decide whether *any* mystery reaction has an orientation requirement.

B5 A single chlorine radical ($Cl^{\bullet}$) needs no particular orientation to react. Suggest why a single atom has no "wrong way" to collide.

B6 Select $H_2 + I_2 \longrightarrow 2\,HI$. This reaction needs a **broadside** hit. In the box, draw the orientation of H_2 relative to I_2 that allows a reaction, then explain why all four atoms must meet at the same instant.

B7 Select $OH^- + CH_3Br \longrightarrow CH_3OH + Br^-$. Turn **ON** "Show reaction conditions" to see the required attack direction.

(a) Draw the successful collision, showing OH^- approaching the carbon from *directly opposite* the bromine atom.

(b) Explain why an attack from the bromine side fails.

Part C

Select $H_2 + Cl_2 \longrightarrow 2\,HCl$. This reaction needs **no** particular orientation, so collision energy is the *only* thing that decides success – ideal for studying E_a. Keep "Show reaction conditions" **OFF**.

C1 Set the energy to each value below and press **Run ×50** each time (press **Clear tally** between trials). Record the effective-collision percentage.

Collision energy	10	30	45	60	90
Effective collisions (%)					

From your data, the lowest energy that produces a reaction is about: ______________. This estimates the E_a of the reaction.

C2 Now turn "Show reaction conditions" **ON** and read the position of the amber E_a marker on the meter. How well does it match your experimental estimate from C1?

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C3 On the axes, **draw and label** the energy profile for this (exothermic) reaction. Label the **reactants**, the **products**, and the **activation energy** (E_a).

C4 Select $N_2 + O_2 \longrightarrow 2\,NO$ and look at its E_a marker (turn conditions ON). It sits extremely high.

(a) What does such a high E_a tell you about the bond that must be broken in N_2?

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(b) Use this to explain why the nitrogen and oxygen in the air around you do not simply react together.

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(c) Name one everyday situation where this reaction *does* occur, and explain what makes it possible there.

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C5 A student writes: "*Heating the reactants lowers the activation energy.*" Use the simulator to decide whether this is correct, and describe what increasing the collision energy actually changes.

Part D

Stay on a reaction of your choice with "Show reaction conditions" ON.

D1 Tick **Add catalyst**. Describe precisely what changes on the energy meter.

D2 Find a collision energy that gives **0%** effective collisions *without* the catalyst but a **higher** percentage *with* it (use **Run ×50** and **Clear tally** to test each case). Record the energy and both percentages, then explain why the same collisions now succeed.

D3 On the axes, draw **two** pathways for the same reaction on one diagram: one without a catalyst and one with. Label both activation energies and show clearly that the catalyst provides a lower barrier.

D4 For each statement, circle **T** or **F** and justify your choice in one line.

(a) A catalyst is consumed by the reaction. **T** / **F**

(b) A catalyst changes the products that are formed. **T** / **F**

(c) A catalyst lowers the activation energy. **T** / **F**

(d) A catalyst makes the reaction release more energy overall. **T** / **F**

Part E

E1 Select $2\,NO + O_2 \longrightarrow 2\,NO_2$ (a **three-body** reaction). At maximum energy, press **Run ×50** and record *Effective collisions*: __________%.
Compare this with the value you found for the two-molecule reaction $CO + NO_2$ in B4: __________%.

E2 A reaction needing three molecules at once is called **termolecular**. Using collision theory and your two percentages, explain why termolecular reactions tend to be so rare – even when there is plenty of energy.

E3 The simulator fires collisions one batch at a time, but a real reaction flask holds around 6×10^{23} molecules colliding constantly. Using collision theory, predict and explain the effect on the **rate of reaction** of:

(a) increasing the **concentration** of the reactants;

(b) increasing the **temperature**.
(Link your answer to what happened to the *effective-collisions* percentage in Part C when you raised the energy.)

E4 A student uses **Run ×50** on two mystery reactions, P and Q, *at the same collision energy*, and gets **30% effective collisions for both**. They conclude: "*P and Q must have identical activation energies.*"

Identify the flaw in this reasoning, and describe how you would use the simulator to actually compare the activation energies of P and Q.